



BIO 004 • HUMAN ANATOMY • SOLANO COMMUNITY COLLEGE

Module 5 Lab Structure List.

Brain • Cord • Peripheral • Autonomic

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How to use this list

This is a quick reference lab study guide. These are the lab structures and concepts to learn for the lab portion of the exam. Refer to the lab worksheets for the study questions that matter for the course. Treat this as the master list. Occasionally a lab sheet will not contain a structure seen here, so always cross reference against this list. Be prepared to identify structures on models, images, the cadaver, and slides from the microscope.

Some items are tagged with the module they are tested in. You meet the slide in an earlier module and are examined on it here.

Each tissue gets its own block. Tick the box by the name once you can identify it on the slide, write in its function and location, then work down the vertical list and tick each structure you can point to. Anything not in a list is context, not something you will be asked to identify.

READ THIS PART TWICE

This is an extensive list. It would be impossible to test on everything, but everything on this list is testable and you should prepare for it. On occasion I may add something to the list or remove something. Make sure you are in class, and have a person to connect with if you miss class, who can update you on any changes.

CHAPTER 1

Nervous Tissue, Histology

THE SLIDES AND THE NEURON MODEL

Five slides, and her master list names all five. Fill in function and location as you go, because on this sheet the cell you are pointing at is often identified by the tissue it sits in rather than by its own shape.

☐ Multipolar neuron, on the slide and the model

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Cell body (soma) | ☐ Cell processes (dendrites) | ☐ Synaptic bulb (terminal end) |
| ☐ Nucleus | ☐ Axon | ☐ Nissl bodies |
| ☐ Nucleolus | ☐ Axon hillock | ☐ Neurofibrils |

☐ Spinal cord, cross section

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|---------------------------------------|--|---|
| ☐ White matter | ☐ Ventral (anterior) horn | ☐ Lateral horn, if present |
| ☐ Gray matter | ☐ Dorsal (posterior) horn | ☐ Central canal |

☐ **Cerebral cortex**

FUNCTION _____

LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Pyramidal cells☐ Gray matter of the cortex☐ Underlying white matter☐ **Cerebellum**

FUNCTION _____

LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Purkinje cells☐ Cerebellar cortex☐ Arbor vitae☐ **Nerve, cross section**

All of these are best seen on a cross section. Work outward from a single axon to the whole nerve.

FUNCTION _____

LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Axon☐ Endoneurium☐ Fascicle☐ Schwann cells☐ Perineurium☐ Nodes of Ranvier☐ Epineurium☐ **The neuroglia**

Sort the six into the four of the central nervous system and the two of the peripheral nervous system, and give the job of each.

STRUCTURES TO IDENTIFY

☐ Astrocytes☐ Microglia☐ Schwann cells☐ Oligodendrocytes☐ Ependymal cells☐ Satellite cells

A cluster of cell bodies is called a _____ in the CNS and a _____ in the PNS.

A bundle of axons is called a _____ in the CNS and a _____ in the PNS.

Which cell builds the myelin in the PNS, and which one builds it in the CNS?

CHAPTER 2**The Brain****THE BRAIN MODEL, THE SHEEP BRAIN, AND THE CADAVER**

The largest identification block of the module. Work it surface first, then midsagittal, then inferior, because that is the order the models are set out in and it is how the tags follow.

☐ **Surface features of the cerebrum****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---|
| ☐ Gyrus | ☐ Precentral gyrus | ☐ Postcentral sulcus |
| ☐ Sulcus | ☐ Precentral sulcus | ☐ Lateral sulcus (Sylvian fissure) |
| ☐ Longitudinal fissure | ☐ Central sulcus | ☐ Insula |
| ☐ Transverse fissure | ☐ Postcentral gyrus | |

☐ **The lobes**

Two lobes are unpaired on her list and two are paired. Say which is which, and name the sulcus or fissure that bounds each lobe.

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---------------------------------|
| ☐ Frontal lobe | ☐ Temporal lobe | ☐ Insula |
| ☐ Parietal lobe | ☐ Occipital lobe | |

☐ **Midsagittal section****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|---|
| ☐ Corpus callosum | ☐ Fornix | ☐ Infundibulum |
| ☐ Sulcus of the corpus callosum | ☐ Thalamus | ☐ Pineal gland |
| ☐ Cingulate gyrus | ☐ Massa intermedia (interthalamic
adhesion) | ☐ Optic chiasm |
| ☐ Cingulate sulcus | ☐ Hypothalamus | ☐ Cerebellum |
| ☐ Septum pellucidum | | ☐ Transverse fissure |

☐ **Inferior surface****STRUCTURES TO IDENTIFY**

- | | | |
|--|---------------------------------------|--|
| ☐ Uncus | ☐ Crus cerebri | ☐ Optic tract |
| ☐ Parahippocampal gyrus | ☐ Optic chiasm | ☐ Olfactory bulb |
| ☐ Mammillary bodies | ☐ Optic nerve | ☐ Olfactory tract |

☐ **Functional areas of the cortex**

Name the gyrus each cortical area sits on, and say which hemisphere Broca's and Wernicke's areas are usually found in.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|-----------------------------------|
| ☐ Primary motor cortex | ☐ Broca's area | ☐ Thalamus |
| ☐ Primary sensory cortex | ☐ Wernicke's area | |

☐ **The cerebellum****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---|
| ☐ Cerebellar hemispheres | ☐ Folia | ☐ Arbor vitae |
| ☐ Vermis | ☐ Cerebellar cortex | ☐ Transverse fissure |

☐ **The cerebral arterial circle (Circle of Willis)**

Trace the ring from a vertebral artery around to an anterior cerebral artery, naming every vessel on the way.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Internal carotid artery | ☐ Posterior cerebral artery | ☐ Anterior inferior cerebellar artery |
| ☐ Anterior cerebral artery | ☐ Basilar artery | ☐ Posterior inferior cerebellar artery |
| ☐ Anterior communicating artery | ☐ Pontine arteries | ☐ Anterior spinal artery |
| ☐ Middle cerebral artery | ☐ Vertebral arteries | |
| ☐ Posterior communicating artery | ☐ Superior cerebellar artery | |

Which gyrus lies immediately anterior to the central sulcus, and what does it do?

CHAPTER 3**The Brainstem and Cerebellum****THE BRAIN MODEL AND THE SHEEP BRAIN**

Three regions from the top down, plus the cerebellum hanging off the back of them. Every structure here is either a surface landmark or a stalk, so name the region before you name the part.

☐ **The three regions**

Name the cranial nerves that arise from each region.

STRUCTURES TO IDENTIFY

- | | |
|-----------------------------------|--|
| ☐ Midbrain | ☐ Medulla oblongata |
| ☐ Pons | ☐ Pontomedullary junction |

☐ **The midbrain****STRUCTURES TO IDENTIFY**

- | | | |
|---|---------------------------------------|--|
| ☐ Cerebral peduncles | ☐ Crus cerebri | ☐ Cerebral aqueduct |
|---|---------------------------------------|--|

☐ **The pons****STRUCTURES TO IDENTIFY**

- | | | |
|---------------------------------------|---|---|
| ☐ Basilar pons | ☐ Middle cerebellar peduncle | ☐ Fourth ventricle |
|---------------------------------------|---|---|

☐ **The medulla oblongata**

Say what crosses at the decussation of the pyramids and what that crossing explains.

STRUCTURES TO IDENTIFY

- | | |
|--|--|
| ☐ Pyramids | ☐ Medullary olive |
| ☐ Decussation of the pyramids | ☐ Pontomedullary junction |

☐ **The cerebellum and its peduncles**

Say which region of the brainstem each peduncle connects to.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Cerebellar hemispheres | ☐ Arbor vitae | ☐ Inferior cerebellar peduncle |
| ☐ Vermis | ☐ Superior cerebellar peduncle | |
| ☐ Folia | ☐ Middle cerebellar peduncle | |

Which region of the brainstem is continuous with the spinal cord, and at what landmark?

CHAPTER 4

Meninges, Ventricles, and CSF

THE BRAIN MODEL AND THE CORONAL SECTION

Three membranes, three spaces, four ventricles, and one circuit. Her master list asks you to trace the fluid from the lateral ventricles to reabsorption, and it warns that some structures on that route are named but not seen on the model.

☐ The cranial meninges

Name the three layers in order from the skull inward.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|------------------------------------|
| ☐ Dura mater | ☐ Meningeal layer of the dura | ☐ Pia mater |
| ☐ Periosteal layer of the dura | ☐ Arachnoid mater | |

☐ The dural extensions

For each one, say where it is found and which structures it separates.

STRUCTURES TO IDENTIFY

- | | | |
|---------------------------------------|---|--|
| ☐ Falx cerebri | ☐ Falx cerebelli | ☐ Tentorium cerebelli |
|---------------------------------------|---|--|

☐ The dural venous sinuses

Say where the sigmoid sinus leaves the skull and what the blood becomes once it does.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Superior sagittal sinus | ☐ Straight sinus | ☐ Transverse sinus |
| ☐ Inferior sagittal sinus | ☐ Confluence of sinuses | ☐ Sigmoid sinus |

☐ The meningeal spaces

Say which of these spaces holds cerebrospinal fluid and which is only a potential space.

STRUCTURES TO IDENTIFY

- | | |
|---|---|
| ☐ Epidural space | ☐ Subarachnoid space |
| ☐ Subdural space | ☐ Arachnoid granulations |

☐ The ventricles

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Lateral ventricle | ☐ Cerebral aqueduct | ☐ Septum pellucidum |
| ☐ Interventricular foramen of Monro | ☐ Fourth ventricle | |
| ☐ Third ventricle | ☐ Central canal | |

☐ **The choroid plexus, on the slide and the model**

FUNCTION _____

LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Choroid plexus☐ Ependymal cells☐ Capillary network

Trace cerebrospinal fluid from the lateral ventricles to its reabsorption, naming every structure it passes. Include the medial and lateral apertures and the central canal even though they are not seen on the model.

CHAPTER 5**The Spinal Cord****THE CORD MODEL, THE CROSS SECTION, AND THE CADAVER**

The cord in two views. The whole cord from the outside gives you the enlargements and the terminations; the cross section gives you the horns, the commissure, and the canal.

☐ **The spinal meninges**

Say what fills the epidural space in the vertebral canal, and why there is no epidural space inside the cranium.

STRUCTURES TO IDENTIFY

☐ Dura mater☐ Epidural space☐ Denticulate ligaments☐ Arachnoid mater☐ Subdural space☐ Pia mater☐ Subarachnoid space☐ **External anatomy of the cord**

Give the vertebral level where the cord ends, and say what fills the canal below that point.

STRUCTURES TO IDENTIFY

☐ Cervical enlargement☐ Cauda equina☐ Posterior median sulcus☐ Lumbar enlargement☐ Filum terminale☐ Conus medullaris☐ Anterior median fissure☐ **The cord in cross section**

Say which segments of the cord carry a lateral horn and what sits in it.

FUNCTION _____

LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Ventral (anterior) horn☐ White matter☐ Anterior column☐ Lateral horn☐ Gray commissure☐ Lateral column☐ Dorsal (posterior) horn☐ Central canal☐ Posterior column☐ Gray matter☐ Anterior white commissure

Where in the vertebral column is a lumbar puncture performed, and why is that level safe?

CHAPTER 6

The Cranial Nerves

THE BRAIN MODEL, THE SKULL, AND THE SHEEP BRAIN

Twelve nerves, and her master list asks four things of each one: identify it, classify it as sensory, motor, or both, give its function, and find the opening it leaves the skull through. Do all four every time you point at one.

☐ The twelve cranial nerves, I to XII

For each nerve give the number, the name, the classification as sensory, motor, or both, and the function.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ CN I, Olfactory | ☐ CN V, Trigeminal | ☐ CN IX, Glossopharyngeal |
| ☐ CN II, Optic | ☐ CN VI, Abducens | ☐ CN X, Vagus |
| ☐ CN III, Oculomotor | ☐ CN VII, Facial | ☐ CN XI, Accessory |
| ☐ CN IV, Trochlear | ☐ CN VIII, Vestibulocochlear | ☐ CN XII, Hypoglossal |

☐ Structures associated with the cranial nerves

Name the nerve that passes through each opening.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Pontomedullary junction | ☐ Olfactory foramina in the cribriform plate | ☐ Optic tract |
| ☐ Internal acoustic meatus | ☐ Optic nerve | ☐ Superior orbital fissure |
| ☐ Olfactory bulb | ☐ Optic chiasm | ☐ Optic canal |
| ☐ Olfactory tract | | |

☐ The branches of the trigeminal nerve

Name the skull opening each division passes through.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Ophthalmic division (V1) | ☐ Maxillary division (V2) | ☐ Mandibular division (V3) |
|---|--|---|

☐ The extrinsic eye muscles

Name the cranial nerve that supplies each muscle. Three nerves cover all seven.

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Lateral rectus | ☐ Inferior rectus | ☐ Levator palpebrae superioris |
| ☐ Medial rectus | ☐ Superior oblique | |
| ☐ Superior rectus | ☐ Inferior oblique | |

Which three cranial nerves are purely sensory? _____

Which four cranial nerves are mixed? _____

CHAPTER 7

Peripheral Nerves and Plexuses

THE CORD MODEL, THE CADAVER, AND THE TORSO

The spinal nerve is the hinge of this sheet. Everything before it is a root, everything after it is a ramus, and her list asks you to say what kind of information travels in each.

☐ **The spinal nerve and its roots**

For each structure say what it carries: sensory axons, motor axons, mixed, or sensory cell bodies.

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Ventral (anterior) root | ☐ Dorsal root ganglion | ☐ Intervertebral foramen |
| ☐ Dorsal (posterior) root | ☐ Spinal nerve | ☐ Rootlets |

☐ **The rami and the sympathetic connections**

Say which ramus forms the plexuses and which one serves the deep muscles and skin of the back.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Ventral (anterior) ramus | ☐ Gray ramus communicans | ☐ Sympathetic chain ganglion |
| ☐ Dorsal (posterior) ramus | ☐ White ramus communicans | ☐ Sympathetic trunk |

☐ **The plexuses**

Give the spinal nerves that form each plexus and the region of the body each one serves.

STRUCTURES TO IDENTIFY

- | | |
|--|--|
| ☐ Cervical plexus | ☐ Lumbar plexus |
| ☐ Brachial plexus | ☐ Sacral plexus |

☐ **Nerves to find on the cadaver and the models**

Name the plexus the phrenic nerve comes from and the structure it supplies.

STRUCTURES TO IDENTIFY

- | | |
|--|---|
| ☐ Phrenic nerve | ☐ Intercostal nerves |
|--|---|

Which root carries a ganglion, and whose cell bodies are in it? _____

A patient loses sensation and strength across one whole upper limb. Which plexus is damaged?
